# Xylocaine 2% MDV

Lidocaine hydrochloride, 50ml solution for injection

**Qualitative and Quantitative Composition**
Lidocaine hydrochloride 20 mg/ml
For excipients see “List of Excipients”.

**Product Description**
Xylocaine solution for injection is a sterile, isotonic aqueous solution. The pH of the solution is 5.0 - 7.0.

**Pharmaceutical Form**
Solution for injection.

**Clinical Particulars**
**Therapeutic indications**
Lidocaine can be used for most major nerve blocks, e.g. sciatic and retrobulbar.

**Posology and method of administration**
**Adults and children above 12 years of age**
Xylocaine should only be used by physicians with experience of regional anaesthesia or under their supervision. The lowest possible dose for adequate anaesthesia should be used.

*Maximum recommended doses:*
Xylocaine 20 mg/ml, 20 ml (400 mg lidocaine hydrochloride)

Table 1 Dosage Recommendations, is a guide to dosage for the more commonly used techniques in the average adult. The clinician's experience and knowledge of the patient's physical status are of importance in calculating the required dose. When prolonged blocks are used, e.g., by repeated administration, the risks of reaching a toxic plasma concentration or inducing a local neural injury must be considered.

**Table 1. Dosage Recommendations**

| **Type of block** | **Conc. mg/ml** | **Dose** | | **Onset (min)** | **Duration (h)** |
|---|---|---|---|---|---|
| **Major nerve block** | | **ml** | **mg** | | |
| Retrobulbar | 20 | 4 | 80 | 3-5 | 1.5-2 |
| Sciatic | 20 | 15-20 | 300-400 | 15-30 | 2-3 |

The doses given in the table are those considered to be necessary to produce successful blocks and should be regarded as a guide for use in adults. Individual variations in onset and duration occur. The figures reflect the expected average dose range needed. Standard textbooks should be consulted for factors affecting specific block techniques and for individual patient requirements.

Unnecessarily high doses of local anaesthetics are to be avoided. In general, surgical anaesthesia requires the use of the higher concentrations and doses. When blocking smaller nerves, or when a less intense block is required, the use of a lower concentration is indicated. The volume of drug used will affect the extent and spread of anaesthesia.

In order to avoid intravascular injection, aspiration should be repeated prior to and during administration of the main dose, which should be injected slowly or in incremental doses, at a rate of 100-200 mg/min, while closely observing the patient's vital functions and maintaining verbal contact. An inadvertent intravascular injection may be recognised by a temporary increase in heart rate and an accidental intrathecal injection by signs of a spinal block. If toxic symptoms occur, the injection should be stopped immediately.

The doses should be reduced for children and patients in reduced general condition.

**Contraindications**

Hypersensitivity to local anaesthetics of the amide type or to other components of the product.

Xylocaine solution for injection with preservatives should not be used in patients who are hypersensitive to methyl and/or propyl parahydroxybensoate (methyl-/propyl paraben), or to their metabolite para amino benzoic acid (PABA).

Xylocaine solution for injection with preservatives should be avoided in patients allergic to ester local anaesthetics that also has the metabolite PABA.

**Special warnings and special precautions for use**

Regional or local anaesthetic procedures, except those of the most trivial nature, should always be carried out in the proximity of resuscitation equipment. Before major blocks, an intravenous cannula should be inserted before the local anaesthetic is injected. Like all local anaesthetic agents, lidocaine can cause acute central nervous and cardiovascular toxic effects in cases of use that leads to high concentrations in the blood. This applies particularly after inadvertent intravascular administration.

Caution should be exercised in patients of the following categories:

- Elderly, and patients with generally reduced general condition
- Patients with degree II or III AV block since local anaesthetics can lower the conduction capacity of the myocardium
- Patients with severe hepatic disease or severely impaired renal function
- Patients treated with class III anti-arrhythmic drugs (e.g. amiodarone) should be closely observed and ECG monitoring should be considered, since the cardiac effects of lidocaine and class III anti-arrhythmic drugs can be additive. (See also "Interaction with other medicinal products and other forms of interaction")

There have been reports of lidocaine-induced hypersensitivity reactions leading to Kounis syndrome (acute allergic coronary arteriospasm that can result in myocardial infarction). Kounis syndrome can develop in patients with and without cardiac risk factors, and may present with cardiac and / or allergic symptoms.

Some regional anaesthesia techniques may be associated with severe adverse reactions, as indicated below:

- Retrobulbar injections can in rare cases reach the cranial subarachnoid space and cause e.g. temporary blindness, cardiovascular collapse, apnoea and convulsions. These symptoms must be treated immediately.
- Retro- and peribulbar injections with local anaesthetics involve some risk of persistent

ocular muscle dysfunction. The primary causes include trauma and/or local toxic effects on muscles and/or nerves. The severity of such tissue reactions is related to the degree of trauma, the concentration of the local anaesthetic, and the duration of exposure of the tissue to the local anaesthetic. For this reason, as with all local anaesthetics, the lowest effective concentration and dose of local anaesthetic should be used. Vasoconstrictors may aggravate tissue reactions and should be used only when indicated.

- There have been post-marketing reports of chondrolysis in patients receiving post- operative intra-articular continuous infusion of local anaesthetics. The majority of reported cases of chondrolysis have involved the shoulder joint. Due to multiple contributing factors and inconsistency in the scientific literature regarding the mechanism of action, causality has not been established. Intra-articular continuous infusion is not an approved indication for Xylocaine.

The prime causes are traumatic nerve injury and/or local toxic effect on muscles and nerves by injected local anaesthetic. The extent of this tissue damage depends on the degree of trauma, the concentration of the local anaesthetic, and the duration of exposure to the local anaesthetic. For this reason, the lowest effective dose should be chosen.

- Inadvertent intravascular injection in the head and neck regions can cause cerebral symptoms even with low doses.

Xylocaine solution for injection is probably porphyrinogenic and should only be prescribed to patients with acute porphyria if there are very strong reasons. Appropriate precautions should be taken for all porphyric patients.

Xylocaine solution for injection, in multidose vials containing methyl parahydroxybenzoate (methylparaben) as preservative and should therefore not be used for anaesthesia by the intrathecal, intracisternal or intra- or retro-bulbar routes.

**Interaction with other medicinal products and other forms of interaction**

Lidocaine should be used with caution in patients receiving other local anaesthetics or agents structurally related to amide-type local anaesthetics e.g. certain anti-arrhythmics, such as mexilitine and tocainide, since the systemic toxic effects are additive. Specific interaction studies with lidocaine and anti-arrhythmic drugs class III (e.g. amiodarone) have not been performed, but caution is advised (See also **Special warnings and special precautions for use**).

Drugs that inhibit~~s~~ the metabolism of lidocaine (e.g. cimetidine) may cause potentially toxic plasma concentrations when lidocaine is given in repeated high doses over a long time period. Such interactions are of no clinical importance following short-term treatment with lidocaine at recommended doses.

Lidocaine should be used with caution with other local anaesthetics or class IB anti-arrhythmic drugs, as the toxic effects are additive.

**Pregnancy and lactation**

*Pregnancy*

Adequate data from treatment of pregnant women is unavailable.

Lidocaine passes the placenta. It is reasonable to assume that a large number of pregnant women and women of child-bearing age have been given lidocaine. No specific disturbances to the reproductive process have so far been reported, e.g., no increased incidence of malformations, or direct or indirect effect on the foetus. However, the risks for human are not completely investigated.

At temporary use during pregnancy and labour the benefit is judged to overweigh the possible risks. Paracervical blockade or pudendal blockade with lidocaine increases the risk for reactions of the foetus, such as bradycardia/tachycardia. Therefore, the heart rate of the foetus must be carefully monitored. See also “Pharmacokinetic properties”.

*Lactation*
Lidocaine is excreted into breastmilk in small amounts. The use of Xylocaine at recommended doses is unlikely to affect the neonate. Breastfeeding can therefore be continued during use of Xylocaine.

**Effects on ability to drive and use machines**
Depending on the dose and method of administration, lidocaine can have a transient effect on movement and coordination.

**Undesirable effects**
Undesirable effects caused by the product itself can be difficult to distinguish from the physiological effects of the nerve block (e.g. fall in blood pressure, bradycardia), events caused directly by the needle puncture (e.g. nerve damage) or caused indirectly by the needle puncture.

| | |
|---|---|
| Very common (>1/10) | *Gastrointestinal disorders:* Nausea.<br>*Vascular disorders:* Hypotension. |
| Common (>1/100, <1/10) | *Cardiac disorders:* Bradycardia<br>*Vascular disorders:* Hypertension.<br>*Nervous system disorders:* Paraesthesia, dizziness.<br>*Gastrointestinal disorders:* Vomiting. |
| Uncommon (>1/1000, <1/100) | *Nervous system disorders:* Symptoms of CNS toxicity (convulsions, circumoral paraesthesia, numbness of the tongue, hyperacusis, visual disturbances, tremor, tinnitus, CNS depression, dysarthria). |
| Rare (>1/10 000, <1/1000) | *Immune system disorders:* Allergic reactions, in the most severe cases anaphylactic shock.<br>*Nervous system disorders:* Neuropathy, peripheral nerve injury, arachnoiditis.<br>*Eyes:* Double vision.<br>*Cardiac disorders:* Cardiac arrest, cardiac arrhythmias.<br>*Respiratory, thoracic and mediastinal disorders:* Respiratory depression. |

**Overdose**

Accidental intravascular injections of local anaesthetics can cause immediate system-toxic reactions (within seconds to a few minutes). Signs of system toxicity in cases of overdose occur later (15-60 minutes after injection) due to a slower increase in the concentration of local anaesthetic in the blood (see "Undesirable effects").

*Toxicity:*
Oral administration: Less than 50 mg seems not to impose any risk for toddlers. 75 mg to a 2 years old child caused a mild, 100 mg to a 5 months old child cause serious, 300 + 300 mg within 4 hours to a 3½ years old child caused serious to very serious, 400-500 mg to a 2 year old child and 1 g during 12 hours to a 1 year old child caused very serious intoxication. 600 mg to an adult caused mild, 2 g to an adult caused moderate intoxication.
Parenteral administration: 50 mg intravenously to a 1 months old child caused very severe intoxication. 200-400 mg infiltration to an adult caused serious, 500 mg to an 80 year old patient and 1 g intravenously to adults caused very severe intoxication.
Topical administration: 8.6-17.2 mg/kg BW to toddlers when used on burn injured skin caused serious intoxication.

*Symptoms:*
First CNS excitation, later CNS depression. When exposed for large doses the first symptom can be rapidly developing seizures. Agitation, dizziness, visual disturbances, perioral paresthesia, nausea. Later ataxia, auditory disturbances, euphoria, confusion, speech difficulties, paleness, sweating, tremors, seizures, coma, apnoea. Different kinds of arrhythmias especially bradycardial arrhythmias but also if exposed to large doses ventricular tachycardia, ventricular fibrillation, QRS prolongation, atrio-ventricular block. Cardiac failure, hypotension. (rare cases of methaemoglobinaemia are described).

*Treatment:*
If oral exposure, active charcoal. (Provoking vomiting may be hazardous due to anaesthesia of the mucous membrane and the risk for seizures in the early phase. If ventricular lavage is indicated, this should be done with a ventricular tube after tracheal intubation.) Seizures are treated with diazepam. Oxygen. If needed tracheal intubation and controlled respiration (if needed hyperventilation). Bradycardia is treated with atropine. Circulatory deterioration is treated with fluids administrated intravenously, dobutamine and if needed epinephrine (initially 0.05 microg/kg BW/min, to be increased if needed by 0.05 microg/kg BW/min each 10 minutes.), in more severe cases guided by haemodynamic monitoring. Ephedrine may also be tried. If circulatory arrest, resuscitation efforts for several hours may be indicated.

**Pharmacological Properties**

**Pharmacodynamic properties**
Pharmacotherapeutic group: Local anaesthetics
ATC code: N01B B02

Xylocaine contains lidocaine, which is a local anaesthetic of the amide type. Lidocaine reversibly blocks impulse conduction in the nerve fibres by inhibiting the transport of sodium ions through the nerve membrane. Similar effects can also be seen on excitatory membranes in the brain and myocardium. Lidocaine has a rapid onset of effect, a high anaesthetic frequency and low toxicity. Lower concentrations of lidocaine have less effect on motor nerve fibres.

**Pharmacokinetic properties**
The rate of absorption is dependent on dose, route of administration and perfusion at the injection site. Intercostal blockades produce the highest plasma concentrations (approx. 1.5 micrograms/ml per 100 mg injected), while subcutaneous injections in the abdomen produce the lowest plasma concentrations (approx. 0.5 micrograms/ml per 100 mg injected). The volume of distribution in a steady state is 91 litres, and plasma protein binding, which is predominantly to alpha-1-acid glycoprotein, is 65 %.

Absorption is total and bi-phasic from the epidural space with half-lives of about 9.3 minutes and 82 minutes. The slow absorption is the time-limiting factor in the elimination of lidocaine, which explains the slower elimination after epidural injection, compared with intravenous injection.

Lidocaine is eliminated predominantly through metabolism. Dealkylation to monoethyl-glycinexylidide (MEGX) is mediated mainly by cytochrome P450 3A4. MEGX is metabolised to 2,6-xylidine and glycinexylidide (GX)- 2,6- dimethylaniline is metabolised further by CYP2A6 to 4-hydroxy-2,6- dimethylaniline, which is the major metabolite in the urine (70 %) and is excreted as conjugate. MEGX has a convulsant activity equivalent to that of lidocaine, while GX is devoid of convulsant activity. MEGX appears to occur in similar plasma concentrations as the parent substance. The elimination rates of lidocaine and MEGX after an intravenous bolus dose are approx. 1.5-2 and 2.5 hours respectively.

On account of the rapid hepatic metabolism, the kinetics are sensitive to all alterations in liver function. The half-life can be more than doubled in patients with impaired liver function. Impaired renal function does not affect the kinetics, but can increase the accumulation of metabolites.

Lidocaine crosses the placenta and the concentration of free lidocaine is the same in the mother and the foetus. However, the total plasma concentration is lower in the foetus, which has a lower degree of protein binding.

## Pharmaceutical Particulars

### List of excipients
*1 ml solution for injection contains:*
Sodium chloride 6 mg (tonicity contributor), methylparahydroxybenzoate 1 mg (preservative), sodium hydroxide/hydrochloric acid (to pH 5.0-7.0), water for injections to 1 ml.

### Incompatibilities
Alkalinisation can cause precipitation as lidocaine is limitedly soluble at pH above 6.5.

### Shelf-life
The expiry date is indicated on the packaging.
The multidose vials should not be used for more than three days after the container has been opened for the first time.

**Special precautions for storage**
Do not store above 30ºC. Do not freeze.

**Nature and contents of container**
*Solution for injection 20 mg/ml*
Multiple dose vial, glass, 5 x 50 ml

**Instructions for use and handling, and for disposal**
Re-sterilisation of Xylocaine is not recommended.

Appropriate control procedures to prevent contamination should be employed, including the following:

- use of single-use sterile injecting equipment;
- use a sterile needle and syringe for each new insertion into the vial;rule out the introduction of contaminated material or fluid into a multidose vial.

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**Manufacturer**
Astrea Monts, 18 Rue de Montbazon, Monts, 37260, France

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